

## Laplace Transform Pairs: Time Response

|                                            | $f(t)$                                                                                                               | $F(s)$                                                                                          |
|--------------------------------------------|----------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------|
| First-order impulse response               | $\frac{K_{dc}}{\tau} (e^{t/\tau})$                                                                                   | $K_{dc} \frac{1}{(\tau s + 1)}$                                                                 |
| First-order step response                  | $K_{dc} (1 - e^{t/\tau})$                                                                                            | $\frac{1}{s} \cdot K_{dc} \frac{1}{(\tau s + 1)}$                                               |
| Second-order, underdamped impulse response | $K_{DC} \left( \frac{\omega_n}{\sqrt{1-\zeta^2}} \right) (e^{-\zeta\omega_n t}) \sin(\omega_d t)$                    | $K_{dc} \frac{\omega_n^2}{(s+\zeta\omega_n)^2 + (\omega_d)^2}$                                  |
| Second-order, underdamped step response    | $K_{DC} \left( 1 - \left( \frac{1}{\sqrt{1-\zeta^2}} \right) (e^{-\zeta\omega_n t}) \cos(\omega_d t + \phi) \right)$ | $\frac{1}{s} \cdot K_{dc} \left( \frac{\omega_n^2}{(s+\zeta\omega_n)^2 + (\omega_d)^2} \right)$ |

$$\omega_d = \omega_n \sqrt{1 - \zeta^2}$$

$$\phi = \tan^{-1}(\zeta / \sqrt{1 - \zeta^2})$$